Adaptive Visual Servoing of UAVs Using a Virtual Camera

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We propose an adaptive image-based visual servoing controller for an unmanned aerial vehicle (UAV) that is based on a virtual camera. The controller regulates relative position and yaw of the vehicle to a planar target consisting of multiple points. We adopt image moment features, which simplify the controller's derivation. The control is adaptive to various system parameters. We prove exponential stability of the error dynamics. Experimental results demonstrate controller performance.

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I. INTRODUCTION

Visual servoing of unmanned aerial vehicles (UAVs) uses computer vision feedback to control vehicle motion. Visual servoing allows a UAV to fly in GPS-denied environments [1] or can be applied to a relative positioning task such as autonomous landing [2]. In this paper we consider visual servoing of a quadrotor UAV equipped with a single downward facing camera mounted on-board. The proposed control enables the vehicle to regulate its relative displacement and yaw to a static visual target located on the ground. We assume a planar target with known orientation. Visual servoing usually is categorized into two approaches: position-based visual servoing (PBVS) and image-based visual servoing (IBVS). A number of researchers have applied PBVS to UAVs, e.g. [2, 3]. In this paper we focus on IBVS methods where the error signal $e_s = s - s^*$ to be stabilized is computed from the difference between a feature vector s, which directly depends on image plane measurements, and it's desired value s^*. Since e_s is computed directly from image plane measurements, we avoid the need for 3-D position reconstruction used in PBVS. Position reconstruction requires accurate camera calibration and a prior 3-D model of the target. In general, IBVS is robust to camera calibration error [4]. Another feature of IBVS is the target object tends not to leave the field of view (FoV) since the design minimizes e_s, which directly depends on image plane measurements.

A number of researchers have contributed to IBVS methods for UAVs. One approach chooses image features which make the system model passive and simplify the controller design [5–8]. This is because the time derivative of the norm of the error function depends only on translational velocity and yaw rate. Popular choices for point features are image moments computed in a spherically projected image plane [5]. However, this feature is insensitive to motion in altitude due to an ill-conditioned interaction matrix. To overcome this, modified spherical image moments were developed [6]. Work similar to [5] for line features is in [7]. Besides spherical image moments, another feature can be obtained from the homography matrix between the desired and current images. This matrix embeds information about the vehicle's relative pose [9]. However, these approaches either control partial motion of the UAV or are limited to a small range of image feature error. A virtual spring approach is proposed in [10, 11] to deal with scenarios where the chosen image feature kinematics do not have passivity properties. This approach assumes the image and target planes remain parallel. This assumption is restrictive since lateral UAV motion leads to a nonparallel image and target plane. A different approach relies on a virtual image plane [8, 12–14]. This plane has zero roll and pitch angle and has the same position and yaw angle of the real camera frame. In this approach, features in the real camera image are converted into the virtual camera image with a geometric transformation which depends on

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estimates of the roll and pitch angles of the camera. These angle measurements are commonly obtained from an attitude and heading reference system (AHRS) which is integrated into most UAV autopilots to stabilize attitude. The virtual camera transforms the system model such that its interaction matrix has a decoupled form which simplifies derivation of the controller. A similar work for linear structure is also proposed in [13]. In [15, 16], a state-transformation approach is proposed which generalizes the virtual camera method mentioned above. Due to the simpler model that results, this paper adopts the virtual camera approach.

The contribution of this paper is to propose an IBVS law which is adaptive to system uncertainties. We consider a simple linear input model for quadrotor thrust. We assume propeller thrust is modelled by the product of a gain and the square of the physical input: pulse-width modulation (PWM) voltage to the electronic speed controller (ESC). We observe from experiments that the value of the thrust gain gradually reduces as the battery voltage drops, and the reduction in thrust has a noticeable effect on UAV motion. Hence, we compensate for variation in this parameter using adaptive control. From our flight tests in [15] we observed that image feature error e_s converges to a nonzero steady state error due to constant measurement bias in attitude. Hence, our design adapts to this measurement error. As in [8] we include adaptation to uncertainty in mass and desired depth. Furthermore, we experimentally validate the performance of the proposed control. This should be compared with work in [8] where only simulation results are provided. To the best of our knowledge, this is the first time an IBVS based on a virtual camera approach and adaptive IBVS have been validated in flight tests.

This paper is organized as follows. Section II describes the camera model, image features, kinematics of the image features, and UAV dynamics. Section III describes the system uncertainty and presents the adaptive IBVS control. The simulation and experimental results are shown in Section IV. Finally, concluding remarks are provided in Section V.

II. IMAGE-BASED VISUAL SERVOING MODELLING

In this section we first introduce reference frames used in the camera model and the UAV dynamics. Next we describe the virtual camera approach, select image moments in the virtual camera image plane as features for visual servoing, and provide the corresponding image feature kinematics. Finally, we specify a rigid body UAV dynamics which is used to derive the IBVS control.

A. Reference Frames

We introduce three reference frames as shown in Fig. 1. A navigation frame $\mathcal{N}$ is fixed to some point on the Earth with basis $\{n_1, n_2, n_3\}$ oriented north, east, and down, respectively. The frame $\mathcal{N}$ is approximately inertial as the Earth rotates slowly. We assume the camera is fixed

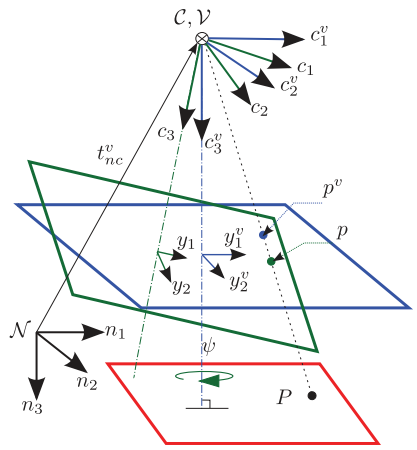

Fig. 1. Pinhole camera geometry model and frame definition.

at the centre of mass (CoM) of the UAV, and define a camera frame $\mathcal{C}$ with basis $\{c_1, c_2, c_3\}$ whose elements are oriented forward, right, and down, respectively. Next, we define a virtual camera frame $\mathcal{V}$ with basis $\{c_1^v, c_2^v, c_3^v\}$ which is attached to the real camera but has zero roll and pitch angles. That is, the virtual camera's image plane remains parallel to the $n_1 - n_2$ plane independent of the vehicle's roll and pitch. The relative orientation of the reference frames can be described by a rotation matrix in $SO(3)$. In this paper we parametrize the rotation between $\mathcal{N}$ and $\mathcal{C}$ with the so-called roll-pitch-yaw (ϕ-θ-ψ) or ZYX Euler angles. Hence, the rotation matrix is given by

$$R = R_{n_3}(\psi)R_{n_2}(\theta)R_{n_1}(\phi) = \begin{bmatrix} c_\psi & -s_\psi & 0 \\ s_\psi & c_\psi & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} c_\theta & 0 & s_\theta \\ 0 & 1 & 0 \\ -s_\theta & 0 & c_\theta \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & c_\phi & -s_\phi \\ 0 & s_\phi & c_\phi \end{bmatrix},$$ where

$c_\xi = \cos\xi$ and $s_\xi = \sin\xi$. We denote $t_{nc}^n = [t_1, t_2, t_3]^T$ as the displacement vector from the origin of $\mathcal{N}$ to the origin of $\mathcal{C}$ expressed in $\mathcal{N}$.

B. Image Feature Selection and Image Kinematics

Visual servoing requires a model of how a camera maps 3-D world coordinates to its 2-D image plane. The camera geometry considered in this paper is based on a commonly used pinhole camera model as shown in Fig. 1. A 3-D point P expressed in the camera frame has coordinates $[X, Y, Z]^T$. The projection of P onto the image plane is a 2-D point feature p with coordinates $p = [y_1, y_2]^T = \lambda[X/Z, Y/Z]^T$, where λ is the focal length of the camera. The kinematics of the image point p is

$$\begin{bmatrix} \dot{y}_1 \\ \dot{y}_2 \end{bmatrix} = \begin{bmatrix} -\frac{\lambda}{Z} & 0 & \frac{y_1}{Z} & \frac{y_1 y_2}{\lambda} & -\frac{y_1^2+\lambda^2}{\lambda} & y_2 \\ 0 & \frac{\lambda}{Z} & \frac{y_2}{Z} & \frac{y_2^2+\lambda^2}{\lambda} & -\frac{y_1 y_2}{\lambda} & -y_1 \end{bmatrix} \nu^c \quad (1)$$

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where $\nu^c = [(v^c)^T, (\omega^c)^T]^T$, and v^c, ω^c denote the linear and angular velocity of the camera expressed in $\mathcal{C}$, respectively. In order to get rid of the roll and pitch dependence in (1) we introduce the notion of a virtual camera. As shown in Fig. 1, the optical centre of the virtual camera is fixed to that of the real camera. The virtual camera moves with the real camera but with zero roll and pitch angle. The point P expressed in $\mathcal{V}$ is $P^v = [X^v, Y^v, Z^v]^T = R_{\theta\phi}P$, $R_{\theta\phi} = R_{n_2}(\theta)R_{n_1}(\phi)$, and the kinematics of P^v is

$$\dot{P}^v = -\left[\dot{\psi}E_3\right]_\times P^v - v^v \tag{2}$$

where $v^v = [v_1^v, v_2^v, v_3^v]^T = R_{\theta\phi}v^c$ and $E_3 = [0, 0, 1]^T$. The corresponding projected image point p^v in the virtual camera image plane is

$$p^v = \left[y_1^v, y_2^v\right]^T = \lambda\left[X^v/Z^v, Y^v/Z^v\right]^T \tag{3}$$

As in [16] the transformation from real camera image coordinates to virtual camera image coordinates can be derived as

$$[y_1^v, y_2^v, \lambda] = \beta\,[y_1, y_2, \lambda]\,R_{\theta\phi}^T$$

where $\beta = \lambda/([y_1, y_2, \lambda]R_{\theta\phi}^T E_3)$. Based on (2) and (3) we obtain the kinematics for the coordinates of a single point in the virtual camera:

$$\begin{bmatrix}\dot{y}_1^v\\ \dot{y}_2^v\end{bmatrix} = \begin{bmatrix}-\lambda/Z^v & 0 & y_1^v/Z^v\\ 0 & -\lambda/Z^v & y_2^v/Z^v\end{bmatrix} v^v + \begin{bmatrix}y_2^v\\ -y_1^v\end{bmatrix}\dot{\psi}$$

REMARK 1 *The virtual camera approach provides a state transformation to remove roll and pitch rate dependence in the kinematics of a point feature's coordinates. This ensures the lateral motion dynamics possess a lower triangular form which simplifies the design of the control law. The transformed lateral dynamics are given below in* (16). *The virtual camera approach provides one method to extend our previous work in [15] which ignores the coupling terms involving* y_1y_2/λ in (1).

Assumption 1 *There are $N > 1$ target points located in a plane parallel to the $n_1 - n_2$ plane.*

This assumption means all the target points expressed in the virtual camera have the same value of depth Z^v. Consequently, we can use the image moments defined in [17, 18] as the image features for visual servoing which leads to a simple interaction matrix [16]. The image features in the virtual camera plane are defined as

$$s_1 = y_{1g}^v s_3 \tag{4a}$$

$$s_2 = y_{2g}^v s_3 \tag{4b}$$

$$s_3 = \sqrt{a^*/a} \tag{4c}$$

where $y_{1g}^v = \sum_{k=1}^N y_{k1}^v$, $y_{2g}^v = \sum_{k=1}^N y_{2k}^v$, $a = \mu_{20} + \mu_{02}$, $\mu_{ij} = \sum_{k=1}^N (y_{1k}^v - y_{1g}^v)^i (y_{2k}^v - y_{2g}^v)^j$, $[y_{1k}^v, y_{2k}^v]^T$ are the virtual image coordinates of the kth point, and a^* is the desired value of a when the vehicle achieves the desired height. The image kinematics can be obtained as

$$\dot{s}_l = -\lambda v_l^v/Z^{v*} - Ss_l\dot{\psi} \tag{5a}$$

$$\dot{s}_3 = -v_3^v Z^{v*} \tag{5b}$$

where $s_l = [s_1, s_2]^T$, $v_l^v = [v_1^v, v_2^v]^T$, $S = \begin{bmatrix}0 & -1\\ 1 & 0\end{bmatrix}$, and $Z_v{}^*$ is the desired depth value of the target plane in $\mathcal{V}$.

REMARK 2 *If the target plane is not horizontal, namely Assumption 1 does not hold, we can still use a virtual camera with zero roll-pitch-yaw rates whose optical axis is parallel to the normal vector of the target plane. The 3-D point expressed in this virtual camera frame is $P^v = R_cRP$, where R_c is the rotation matrix from the target plane to $\mathcal{N}$. The matrix R_c can be obtained from image points as pointed out in [19]. The virtual image coordinates can be obtained with a geometric argument as*

$$\left[y_1^v, y_2^v, \lambda\right] = [y_1, y_2, \lambda]\,\beta R^T R_c^T$$

and its kinematics is

$$\begin{bmatrix}\dot{y}_1^v\\ \dot{y}_2^v\end{bmatrix} = \begin{bmatrix}-\lambda/Z^v & 0 & y_1^v/Z^v\\ 0 & -\lambda/Z^v & y_2^v/Z^v\end{bmatrix} v^v$$

Based on the projected image points, visual seroving can be achieved using various features such as image distance, moments, and area. Using features defined in (4) *the kinematics is*

$$\begin{bmatrix}\dot{s}_1\\ \dot{s}_2\end{bmatrix} = -\frac{\lambda}{Z^{v*}}\begin{bmatrix}v_1^v\\ v_2^v\end{bmatrix}$$

$$\dot{s}_3 = -v_3^v/Z^{v*}$$

REMARK 3 *The virtual camera approach needs at least two points to control the translational motion of the vehicle. It is similar to the spherical projection-based method in [19] but is more general in the number of target points required and how they are configured. In [19], points are projected on a plane parallel to the target plane and tangent to the spherical projection surface. The centroid on that plane and the area of the convex hull of the projected points are required for the image feature definition. Hence, at least three noncollinear target points are needed in [19].*

The features defined in (4) are selected to control the translational motion of the vehicle. As in [17] a fourth feature s_4 is selected to control the yaw motion of the vehicle:

$$s_4 = \arctan\left(2\mu_{11}/\left(\mu_{20} - \mu_{02}\right)\right)/2$$

The kinematics of this feature is

$$\dot{s}_4 = -\dot{\psi} \tag{7}$$

The control objective is to regulate the image feature $s = [s_1, s_2, s_3, s_4]^T$ to a desired constant value s^*. Without loss of generality we take the setpoint as $s^* = [0, 0, 1, 0]^T$.

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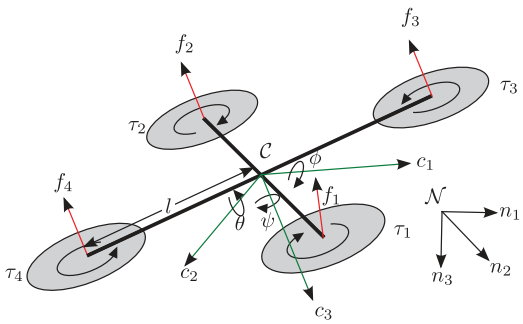

Fig. 2. Quadrotor modelling.

C. Quadrotor Modelling

We write the quadrotor's translational dynamics in $\mathcal{V}$ and the rotational dynamics in $\mathcal{C}$ as

$$\dot{v}^v = -\left[\dot{\psi} E_3\right]_\times v^v + g E_3 + F^v/m \tag{8a}$$

$$\dot{R} = R\left[\omega^c\right]_\times \tag{8b}$$

$$\dot{\omega}^c = -J^{-1}\left[\omega^c\right]_\times J\omega^c + J^{-1}\tau^c \tag{8c}$$

where g is the gravity constant, m is the mass of the vehicle, J is the inertia of the vehicle, $[x]_\times\ y = x \times y$, F^v is the external force acting on the vehicle expressed in $\mathcal{V}$, and τ^c is the external torque acting on the vehicle expressed in $\mathcal{C}$. The quadrotor model includes a number of simplifications such as no small body forces, no gyroscopic effects, and negligible motor system inertia. We approximate F^v as

$$F^v = -R_{\theta\phi} E_3 \sum_{i=1}^{4} f_i \tag{9}$$

where f_i is the thrust created by the ith propeller as shown in Fig. 2. We take

$$f_i = K_T \tilde{W}_i^2$$

where $\tilde{W}_i = W_i - W_{\min}$, W_i is the PWM signal of the ith motor, $W_{\min}$ is the minimum pulse width to start the motor, and K_T is the thrust coefficient. The torque τ^c is approximated as

$$\tau^c = \begin{bmatrix} l\,(f_2 + f_3 - f_1 - f_4)/\sqrt{2} \\ l\,(f_1 + f_3 - f_2 - f_4)/\sqrt{2} \\ K_\tau\left(-\tilde{W}_1^2 - \tilde{W}_2^2 + \tilde{W}_3^2 + \tilde{W}_4^2\right) \end{bmatrix}$$

where l is the distance from the motor to the CoM of the vehicle, and K_τ is a torque constant.

III. ADAPTIVE CONTROLLER DESIGN

In this section we motivate the use of an inner-outer loop control structure for visual servoing and describe the system uncertainty accounted for in the adaptive control. We present the controller design and prove the asymptotic stability of the closed loop.

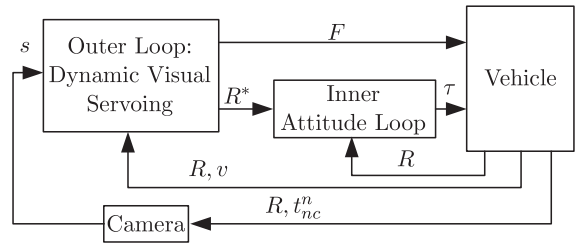

Fig. 3. Visual servoing with inner-outer control loop structure.

A. Inner-Outer Loop Control Structure

The control of UAVs is typically divided into an inner and outer loop structure as shown in Fig. 3. This two-loop structure is used in this paper and is often used independent of the sensors employed (e.g. camera or GPS). The inner loop tracks an orientation reference of the vehicle, and the outer loop provides a roll, pitch, and yaw reference to the inner loop in order to achieve control of translational and yaw motion. The two loop structure arises in part from a difference in sensor sample rates. It also provides a reliable solution for motion control found in many autopilots [20, 21]. In the case of visual servoing, the decoupled control structure is natural due to the low frequency at which the camera acquires and processes images. On the other hand, the relatively fast inertial measurement unit (IMU) sensors driving the inner attitude control allow it to run at a higher rate. A theoretical analysis of the stability of this closed-loop structure is in [16, 22].

B. Uncertainty Modelling

From our previous experimental results in [15] we observed that lateral image feature error converged to a nonzero constant. As confirmed by the work in [16], the bias errors in Euler angle estimates from the AHRS are the cause of this image feature error. To account for this error we model $R_{\theta\phi}$ as $R_{\theta\phi} = R_{n_1}^T(\phi_e)R_{n_2}^T(\theta_e)R_{n_2}(\theta_m)R_{n_1}(\phi_m)$, where the bias errors ϕ_e, θ_e are taken as constants, and the ϕ_m, θ_m are measured roll and pitch angles. The errors ϕ_e and θ_e are treated as additive input disturbances to the outer loop and compensated with the adaptive control. Although we treat ϕ_e and θ_e as input disturbances in the force model, we assume the small biases have negligible influence on the kinematics of the image moments (5). This assumption is practical and made to simplify the derivation of the control law. More details on the justification for this assumption can be found in [16].

During system identification of the UAV model we observed the thrust constant K_T slowly decreases with time as the battery voltage drops during a flight. This is shown in Fig. 4 for one Turnigy 1100 KV brushless motor attached to an 11 × 4.5" propeller. The motor is powered by two 3 cell 2600 mAh LiPo batteries. To compensate for its variation we treat K_T as an unknown constant parameter. The control objective in this paper is to regulate the image feature s to its desired value s^* while adapting to

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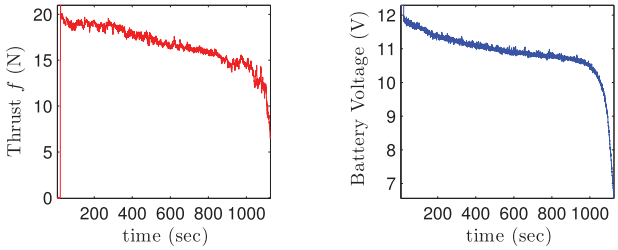

Fig. 4. Time evolution of thrust. PWM signal is set to 1.4 ms.

uncertainty in mass m, thrust constant K_T, desired depth Z^{v*},and the constant biases ϕ_e and θ_e that appear in force model F^v.

C. Outer-Loop Controller Design

In our previous work [15, 16] we observed that reference roll-pitch angles have to be limited to a small range to avoid the target leaving the camera's FoV. Given the small range of roll and pitch we make a small angle approximation so that the force model in (9) becomes

$$F^v = R_{n_1}^T(\phi_e)R_{n_2}^T(\theta_e)R_{n_2}(\theta_m)R_{n_1}(\phi_m)E_3\sum_{i=1}^{4} f_i$$
$$\approx K_T u \begin{bmatrix} -\theta_m + \theta_e \\ \phi_m - \phi_e \\ -1 \end{bmatrix} \tag{10}$$

where $u = \sum_{i=1}^{4} \tilde{W}_i^2$. Based on (5b), (8a), and (10) we write the height subsystem as

$$\dot{s}_3 = -v_3^v/Z^{v*} \tag{11a}$$

$$\dot{v}_3^v = g - K_T u/m \tag{11b}$$

In this subsystem the control input is chosen as the thrust u and the control law is designed as

$$u = K_{h2}\delta_{h2} + \hat{C}_g \tag{12}$$

with

$$\dot{\hat{C}}_g = K_{hg}\delta_{h2} \tag{13}$$

$$\delta_{h1} = s_3 - 1 \tag{14}$$

$$\delta_{h2} = v_3^v/K_{h1} - \delta_{h1} \tag{15}$$

where K_{h1}, K_{h2}, K_{hg} are positive control gains, and $\hat{C}_g$ is the estimate of $C_g = mg/K_T$. From (5a), (8a), and (10) the lateral subsystem dynamics is

$$\dot{s}_l = -\lambda v_l^v/Z^{v*} - Ss_l\dot{\psi} \tag{16a}$$

$$\dot{v}_l^v = K_T u S(\eta_{m1} - \eta_{e1})/m - Sv_l^v\dot{\psi} \tag{16b}$$

where $\eta_{m1} = [\phi_m, \theta_m]^T$, and $\eta_{e1} = [\phi_e, \theta_e]^T$. In this subsystem we choose η_{m1} as the input and the control is taken as

$$uS\eta_{m1} = uS\hat{\eta}_{e1} - K_{l2}\delta_{l2} \tag{17}$$

with

$$\dot{\hat{\eta}}_{e1} = uK_{lrp}S\delta_{l2} \tag{18}$$

$$\delta_{l1} = s_l \tag{19}$$

$$\delta_{l2} = v_l^v/K_{l1} - \delta_{l1} \tag{20}$$

where K_{l1}, K_{l2}, K_{lrp} are positive gains, $\hat{\eta}_{e1} = [\hat{\phi}_e, \hat{\theta}_e]^T$, and $\hat{\phi}_e$ and $\hat{\theta}_e$ are the estimates of ϕ_e and θ_e, respectively.

In order to make s_4 converge to zero the yaw rate reference is set as

$$\psi = K_\psi \int_0^t s_4(\xi)d\xi \tag{21}$$

where $K_\psi > 0$.

THEOREM 1 *The equilibrium* $[\delta_{h1}, \delta_{h2}, \delta_{l1}, \delta_{l2}, s_4]^T = 0$ *of the system (7), (11), (16), with control law (12), (17), (21) and parameter update laws (13), (18), is exponentially stable if all control gains are positive and satisfy*

$$K_{h2} > C_z K_{h1}^2 \tag{22a}$$

$$K_{l2} > \lambda C_z K_{l1}^2 \tag{22b}$$

PROOF Using (7) and (21), we obtain the closed-loop system for yaw motion as

$$\dot{s}_4 = -K_\psi s_4$$

which is globally exponentially stable.

With (11) and the definition of δ_{h1} and δ_{h2} in (14) and (15), we have

$$\dot{\delta}_{h1} = -\frac{K_{h1}}{Z^{v*}}(\delta_{h1} + \delta_{h2}) \tag{23}$$

$$\dot{\delta}_{h2} = -\frac{K_T}{mK_{h1}}\left(u - C_g\right) + \frac{K_{h1}}{Z^{v*}}(\delta_{h1} + \delta_{h2}) \tag{24}$$

Using (23), (24), (13), and (12), we have

$$\dot{x}_h = A_h x_h \tag{25}$$

where $x_h = [\delta_{h1}, \delta_{h2}, \tilde{C}_g]^T$, $\tilde{C}_g = \hat{C}_g - C_g$, $A_h = \begin{bmatrix} -\frac{K_{h1}}{Z^{v*}} & -\frac{K_{h1}}{Z^{v*}} & 0 \\ \frac{K_{h1}}{Z^{v*}} & -\frac{1}{Z^{v*}}(\frac{K_{h2}}{C_z K_{h1}} - K_{h1}) & -\frac{1}{C_z Z^{v*} K_{h1}} \\ 0 & K_{hg} & 0 \end{bmatrix}$, and $C_z = m/(Z^{v*}K_T)$. Using the Routh-Hurwitz stability criterion we can conclude that when K_{h1}, K_{h2}, K_{hg} are positive and (22a) is satisfied, the origin of the system (25) is exponentially stable.

From the definition of δ_{l1} and δ_{l2} in (19) and (20), and the lateral subsystem (16), we obtain

$$\dot{\delta}_{l1} = -\frac{\lambda K_{l1}}{Z^{v*}}(\delta_{l1} + \delta_{l2}) - \dot{\psi}S\delta_{l1} \tag{26}$$

$$\dot{\delta}_{l2} = \frac{K_T u}{mK_{l1}}S(\eta_{m1} - \eta_{e1}) - \dot{\psi}S\delta_{l2} + \frac{\lambda K_{l1}}{Z^{v*}}(\delta_{l1} + \delta_{l2}) \tag{27}$$

Using (26), (27), (18), and (17), we have

$$\dot{x}_l = A_l x_l + D(t)x_l \tag{28}$$

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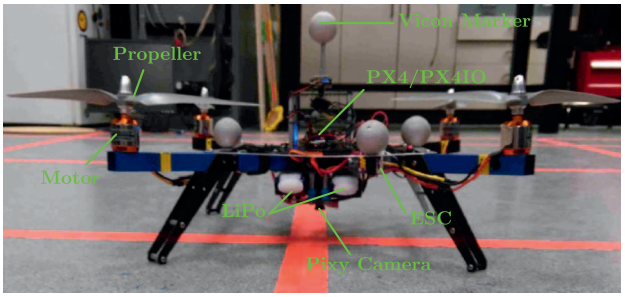

Fig. 5. ANCL quadrotor platform.

where

$$A_l = \begin{bmatrix} -\frac{\lambda K_{l1}}{Z^{v*}} I_2 & -\frac{\lambda K_{l1}}{Z^{v*}} I_2 & 0_{2\times 2} \\ \frac{\lambda K_{l1}}{Z^{v*}} I_2 & -\frac{1}{Z^{v*}}\left(\frac{K_{l2}}{C_z K_{l1}} - \lambda K_{l1}\right) I_2 & \frac{u_g S}{C_z Z^{v*} K_{l1}} \\ 0_{2\times 2} & u_g K_{lrp} S & 0_{2\times 2} \end{bmatrix},$$

$$D(t) = \begin{bmatrix} -\dot{\psi} S & 0_{2\times 2} & 0_{2\times 2} \\ 0_{2\times 2} & -\dot{\psi} S & \frac{\tilde{u}}{C_z Z^{v*} K_{l1}} \\ 0_{2\times 2} & \tilde{u} K_{lrp} S & 0_{2\times 2} \end{bmatrix}, x_l = [\delta_{l1}^T, \delta_{l2}^T, \tilde{\eta}_{e1}^T]^T,$$

$\tilde{u} = u - u_g$, and u_g is the thrust input to compensate gravity. Using the Routh-Hurwitz stability criterion we can conclude that the origin of the nominal system $\dot{x}_l = A_l x_l$ is exponentially stable if K_{l1}, K_{l2}, K_{lrp} are positive and (22b) is satisfied. Since u and $\dot{\psi}$ will exponentially converge to u_g and 0, respectively, we can conclude that $D(t)$ exponentially converges to 0. Using a converse Lyapunov theorem [23, Thm. 4.14] we can show the origin of the system (28) is exponentially stable.

IV. EXPERIMENTAL RESULTS

In this section, we describe our experimental platform and provide its model parameters. We also provide experimental validation of the proposed adaptive control.

A. Experimental Platform Description

To research nonlinear control and visual servoing of UAVs we developed the indoor Applied Nonlinear Control Lab (ANCL) quadrotor platform. The quadrotor vehicle used in this platform is shown in Fig. 5. The quadrotor has a 3DRobotics vehicle frame. The autopilot system is based on PX4 hardware which is an open-hardware and open-source project [21]. In particular, we use the PX4FMU flight management board and the PX4IO interface board. The PX4FMU contains a 168 MHz ARM processor, a 3-D accelerometer, a 3-D gyroscope, and a 3-D magnetometer. An extended Kalman filter-based AHRS fuses the PX4FMU sensor data to provide an estimate of attitude and angular velocity. The PX4FMU computes the adaptive control law and generates the PWM signals to four ESCs which power four 1100 KV brushless motors with 11 × 4.5" multirotor propellers. The on-board computer vision system is a CMUcam5 Pixy which is a low cost vision sensor. It has a nominal focal length of $\lambda = 2.8$ mm, an image sensor size of 1/4", and a maximum resolution of 1280 × 800 pixels. The lens mounted on the Pixy has 75° horizontal and 47° vertical FoV. The Pixy camera can find hundreds of objects in each frame at up to 50 frames per second. The Pixy camera is mounted on the bottom of the quadrotor and connected to the PX4 via a serial connection. It transmits image point features at a rate of about 25 Hz based on a 320 × 200 pixel image. An eight camera Bonita Vicon motion capture system measures the vehicle pose at about 200 Hz with millimetre accuracy. A PC runs the Vicon Tracker software which computes the vehicle pose. It also runs a program to estimate the vehicle's translational velocity. The PX4FMU communicates with this PC via an on-board 2.4 GHz radio to obtain the translational velocity feedback. Power is supplied by two 11.1 V 2600 mAh LiPo batteries. The entire vehicle system weight including battery and computer vision system is 1.6 kg. Table I summarizes the quadrotor and camera model parameters.

TABLE I
Quadrotor and Camera Model Parameters

m	1.6 kg
J	diag([0.03, 0.03, 0.05]) kg·m^2
λ	2.8 mm
Image Sensor Size	1/4"
Image Resolution	320 × 200 pixels
ℓ	0.25 m

TABLE II
Outer Loop Control Gains

Gain	K_{h1}	K_{h2}	K_{hg}	K_{l1}	K_{l2}	K_{lrp}	K_ψ	K_z
Value	0.9	0.09	0.04	0.3182	0.07	0.02	0.5	0.01

TABLE III
PID Control Gains for Inner Loop Attitude Tracking

	Roll	Pitch	Yaw
K_p	0.33	0.345	0.3
K_d	0.08	0.09	0.08
K_i	0.03	0.03	0.005

B. Experimental Results

The visual servoing objective is to make the vehicle hover at a constant position and yaw relative to the target of interest by regulating the image feature error. We use two target points separated by 0.36 m and located on the horizontal ground. The control gains of the outer loop are given in Table II. A proportional-integral-derivative (PID) control is used for the inner loop with gains given in Table III. A block diagram of the implementation is shown in Fig. 6, where the mixer block distributes PWM to the individual ESCs based on u and τ^c. To verify the effectiveness of the adaptive law to compensate the unknown measurement bias and variation in the thrust coefficient K_T, we consider two cases in experiment: with and without the adaptive law (13) and (18). In the adaptive

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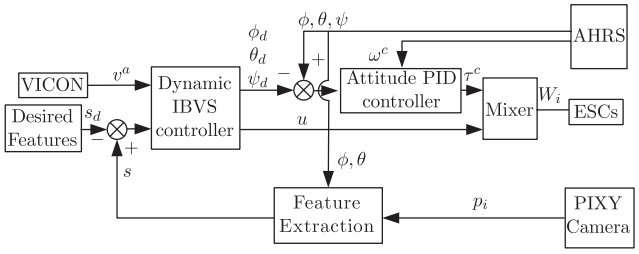
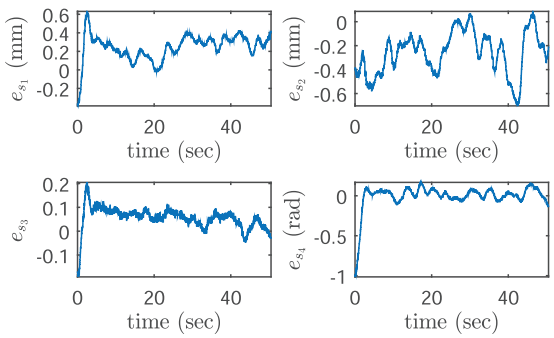
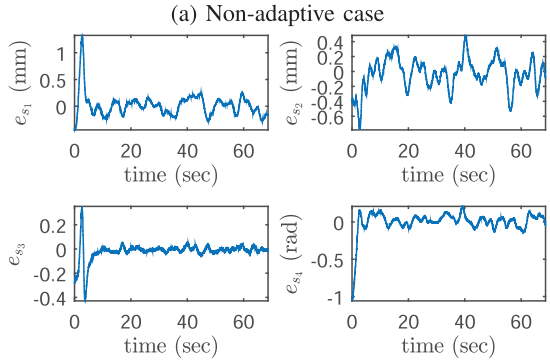


Fig. 6. Block diagram of adaptive visual servoing with inner-outer control loop structure.



(a) Non-adaptive case



(b) Adaptive case

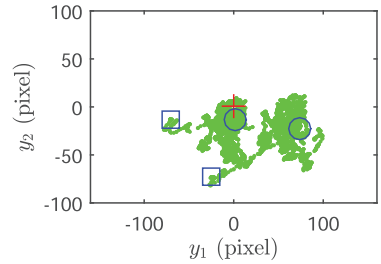
Fig. 7. Experimental trajectories of image feature error e_s.

case, the initial values of $\hat{\phi}_e$, $\hat{\theta}_e$ are set to zero. The estimated value of C_g in the nonadaptive case is 0.435, which is obtained from a manual flight test, and this value is the initial value of $\hat{C}_g$ in the adaptive case.

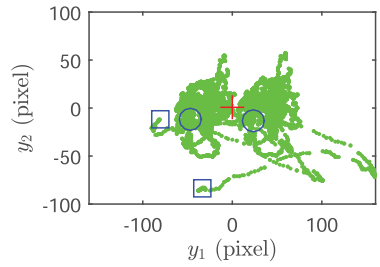
The image feature error is denoted $e_s = [e_{s_1}, e_{s_2}, e_{s_3}, e_{s_3}]^T = s - s^*$. The trajectory of the image feature error e_s is given in Fig. 7. The trajectories of two image points are shown in Fig. 8. The desired 3-D translational displacement of the UAV is denoted t_{nc}^{n*} and translational error is

$$e_{t_{nc}^n} = \left[e_{t_1}, e_{t_2}, e_{t_3}\right]^T = t_{nc}^n - t_{nc}^{n*}$$

where $t_{nc}^{n*} = [0.050, -0.157, -1.337]^T$ m. This value of t_{nc}^{n*} corresponds to $e_s = 0$. The desired yaw of the UAV is $\psi^* = -0.221$ rad, and yaw tracking error is $e_\psi = \psi - \psi^*$. Fig. 9 shows the corresponding 3-D translational error $e_{t_{nc}^n}$ and yaw error e_ψ. Typical inner-loop tracking performance is shown in Fig. 10. Fig. 11 illustrates the parameter estimates $\hat{\phi}_e$, $\hat{\theta}_e$, and $\hat{C}_g$. Based on these plots, both cases reach steady state in about 15 s. Table IV gives

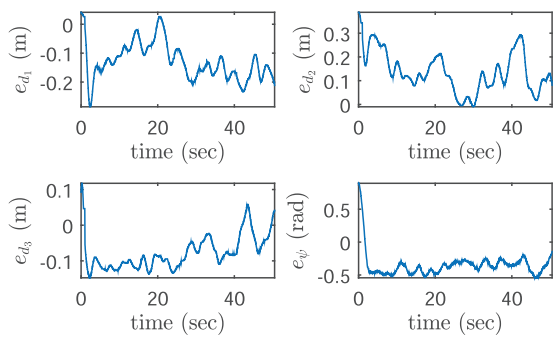


(a) Non-adaptive case

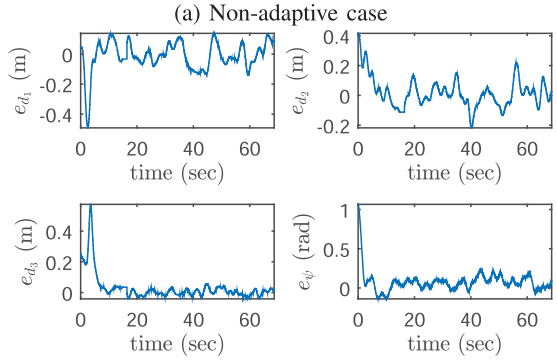


(b) Adaptive case

Fig. 8. Image point trajectories. Initial conditions are shown with squares, circles denote final points and plus symbol represents centre of image. (a) Nonadaptive case. (b) Adaptive case.



(a) Non-adaptive case



(b) Adaptive case

Fig. 9. Experimental trajectories of vehicle's 3-D position error $e_{t_{nc}^n}$ and e_ψ. (a) Nonadaptive case. (b) Adaptive case.

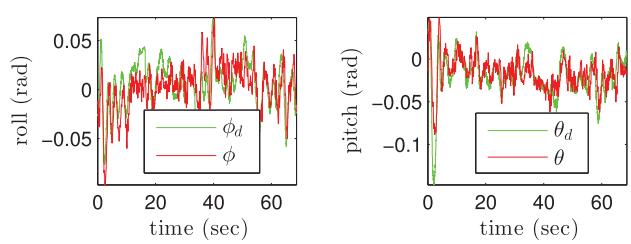


Fig. 10. Example of roll and pitch tracking performance.

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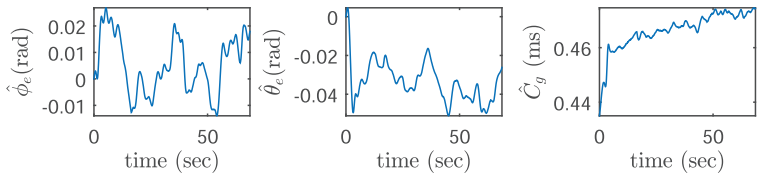

Fig. 11. Experimental trajectories of estimated parameters $\hat{\phi}_e$, $\hat{\theta}_e$, and $\hat{C}_g$.

TABLE IV
Statistics of e_s, $e_{t_{nc}^n}$, and e_ψ.

	Adaptive Control		Nonadaptive Control	
Variable	μ	σ	μ	σ
e_{s_1}	0.00	0.13	0.25	0.10
e_{s_2}	−0.01	0.18	−0.23	0.17
e_{s_3}	−0.00	0.02	0.05	0.03
e_{s_4}	0.02	0.07	0.02	0.06
e_{t_1}	−0.008	0.072	−0.138	0.059
e_{t_2}	−0.002	0.077	0.116	0.070
e_{t_3}	0.004	0.021	−0.074	0.044
e_ψ	0.070	0.061	−0.108	0.082

Note: We denote mean as μ and standard deviation as σ.

the mean and standard deviation of e_s, $e_{t_{nc}^c}$, and e_ψ in steady state. It can be seen from Fig. 7 that e_{s_3} in the nonadaptive case eventually decreases slowly due to the slowly reducing battery voltage. In the adaptive case the value of e_{s_3} stays around 0. We also observe from Table IV that in the adaptive case the mean value of e_{s_1} and e_{s_2} are significantly reduced with similar standard deviations. The trajectories of e_{s_4} converge to zero for both cases, and the performance for regulating the yaw motion in the two cases is similar. This is to be expected since the same control law (21) is used. The trajectories of points in the image plane shown in Fig. 8 are consistent with Table IV. That is, the lower mean values of e_{s_1}, e_{s_2} correspond to a more centered steady state image. We remark that in Fig. 8, the image point trajectories in the adaptive case are less concentrated than the nonadaptive case. This is because the height of the vehicle is higher that that of the adaptive case. This means the same lateral motion of the vehicle for the nonadaptive case creates less motion in the image plane. Fig. 9 and Table IV show the 3-D position errors in the adaptive case are much smaller than the nonadaptive case. Hence, we conclude that the proposed control provides improved visual servoing motion control.

V. CONCLUSION

We have presented an adaptive IBVS control that uses image moment features from projected points in a virtual camera image. The image moment features in the virtual camera image lead to a simple interaction matrix. The proof of exponential stability of the error dynamics is given for the case of target points in a horizontal plane. The proposed control does not require an estimate of depth, known mass, and thrust coefficient. The method is robust to measurement bias in roll and pitch. An indoor quadrotor platform is used to validate the control. The results demonstrate improved performance with lower mean image and position error. We explain how the proposed control can be extended to the nonhorizontal target case for motion control in the translational degrees of freedom. Future work involves developing an output feedback controller to remove the requirement of velocity measurement. Also, achieving regulation of yaw for the nonhorizontal target plane case is planned.

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